MFASHA: A Kinyarwanda Agriculture Chatbot with Text and Voice Support

Technical Report: Final Version of the Product

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Repository: https://github.com/carine-ahishakiye/Mfasha-Chatbot-.git

· Walkthrough video: link in the repository README (section 9)

1. Scope alignment and feature validation

MFASHA implements the eight-stage pipeline of the approved proposal as a FastAPI service with a browser front end. Answers are retrieved from a reviewed knowledge base, never generated, and questions the system is unsure about are referred to an agricultural officer. Table 1 maps each planned feature to its status.

Status key: Done = implemented and evidenced. Partial = implemented with a gap. Changed = delivered differently from the proposal. No evidence = nothing in the repository supports it.

Planned feature	Intended purpose	Status	Evidence (repository / notebook)
Text question answering	Farmer asks in Kinyarwanda and reads a written answer	Done	POST /api/ask/text (api/routes.py); frontend/chat.js; tests/integration/test_api.py
Voice question recording	Spoken input for farmers who find typing hard	Done	chat.js records with MediaRecorder, converts to 16 kHz mono WAV in the browser, 30 s cap; POST /api/ask/voice
Speech-to-text	Turn the recording into text	Done	MMS-1b-all (kin adapter) fine-tuned on 2,000 clips (2.4 h); test WER 0.301 → 0.277; speech/stt.py; results/asr_finetuned_test.csv
Text preprocessing	Normalise spacing, apostrophes,	Done	nlp/preprocessing.py; tests/unit/test_preprocessing.py

Planned feature	Intended purpose	Status	Evidence (repository / notebook)
	greetings before search		
Question classification (KinyaBERT vs TF-IDF + LR/SVM)	Narrow the search and detect off-topic questions	Partial	TF-IDF + LR/SVM trained (test macro-F1 0.89 / 0.93 on seen passages). KinyaBERT failed to load in the notebook (cell 5), so no comparison. Labels are keyword drafts; the officer-category column is empty
Retrieval and ranking (kiny-colbert-free vs BM25 / TF-IDF)	Find the best verified passage	Changed	BM25, TF-IDF, mpnet, e5, bge-m3 and LaBSE compared; fine-tuned multilingual-e5-large adopted (R@1 0.87–0.89). kiny-colbert-free was audited, not benchmarked; results/retrieval_final_test_all_systems.csv
Referral to an officer	Avoid unsafe or invented advice	Done	referral/policy.py (4 chained rules); 40 of 40 out-of-scope questions referred; results/oos_results.csv
Verified knowledge base	Serve only checked answers	Partial	941 passages from kinya-ag-retrieval, all marked reviewed (27–30 Oct 2026, one reviewer); no approved external documents added; results/review_sheet_all_passages.xlsx
Text-to-speech (kinya-flex-tts)	Read the answer aloud	Changed	facebook/mms-tts-kin used: stock voice beat the fine-tuned one (round-trip WER 0.454 vs 0.761). kinya-flex-tts audited only; native-speaker ratings not in the repository; speech/tts.py; results/tts_choice.json
Farmer interface	Simple UI for users with limited digital skills	Changed	HTML/CSS/JS served by FastAPI instead of Streamlit: landing page, chat, history, confidence badge, read aloud, feedback (frontend/)
Storage of questions, answers, feedback	Data for evaluation	Changed	SQLAlchemy tables questions, answers, feedback, expert_reviews; SQLite by default, PostgreSQL via docker-compose; simplified from the 10-entity ERD (storage/orm.py)
Officer review and admin functions	Officers rate answers; admins monitor	Partial	API-key endpoints POST /api/officer/reviews and GET /api/admin/stats; no web screen and no user management

Planned feature	Intended purpose	Status	Evidence (repository / notebook)
Response time (5 s target)	Usable speed	Done (text)	p95 55 ms (hybrid) and 1.05 s (embedding) on 941 passages; STT/TTS latency not benchmarked; benchmark_*.json
Validation interviews and pilot (about 30 farmers, 3 officers, ≥70% correct)	Confirm need; measure usefulness	No yet	No interview notes or pilot data in the repository. The stats and officer-review endpoints exist to collect them

Table 1. Planned features, status and evidence.

Scope changes and justification

The proposal treated KinyaBERT, kiny-colbert-free and kinya-flex-tts as candidates, to be adopted only if they clearly beat a baseline. All three were audited from their model cards (results/candidate_audit.json). KinyaBERT could not be loaded and the other two were not benchmarked, so the delivered system uses components that were measured:

- **Classification:** TF-IDF classifiers in place of KinyaBERT.
- **Retrieval:** fine-tuned e5, adopted because its gain over BM25 has a 95% interval of +0.36 to +0.44 R@1.
- **Voice output:** the stock MMS voice.
- **Interface:** served by the same FastAPI app, so the product deploys as one service and the browser handles audio capture and WAV conversion itself.
- **Database:** simplified, with transcription on the question row and component names on the answer row (storage/orm.py).

2. Technical functionality

Request path

Figure 1 is implemented in src/mfasha/pipeline.py. Typed questions start at stage 3. The referral policy (stage 6) decides whether stage 7 returns an answer or a referral message.

Stage	Component	Detail
1	Voice input	Browser, 16 kHz WAV
2	Speech-to-text	MMS-1b-all + kin adapter
3	Preprocessing	TextCleaner
4	Classification	TF-IDF + SVM / LR

Stage	Component	Detail
5	Retrieval	BM25, TF-IDF, hybrid, e5
6	Referral policy	4 chained rules
7	Verified answer	Reviewed passage + source
8	Text-to-speech	MMS-TTS-kin

Figure 1. Request path (shown as a table).

2.1 Front end and back end

The browser (frontend/chat.js) calls /api/ask/text, /api/ask/voice (multipart WAV) and /api/feedback. Each response carries the answer, status (answered or referred), referral reason, retrieval score, review status, source, per-stage timings and optional base64 audio.

The page shows a confidence badge from the retrieval score (≥0.60 high, ≥0.35 medium, otherwise low), read-aloud, copy and thumbs up/down buttons, and a conversation history kept only in the browser's localStorage. Server text is written with textContent; the only two innerHTML uses are static SVG icons. Controls are labelled and the chat log is an aria-live region.

Every interaction is stored with the components used and stage timings, and /api/admin/stats reports referral rate, latency, ratings and the officer-correct rate (objective 3).

2.2 Data handling and core logic

- **Knowledge base.** C4IR-RW/kinya-ag-retrieval (CC-BY-4.0) is cleaned by scripts/prepare_data.py (spacing and apostrophe fixes, heading-only passages dropped) into 941 passages. Each passage gets a draft category from a keyword labeller (8 categories; "markets and inputs" had only 13 passages and was merged into general farming in the notebook) and a reviewed flag read from the officer sheet.
- **Retrieval.** BM25 is written from scratch over an inverted index, so a search touches only passages that contain a query term, and scores are normalised to 0–1 so one threshold means the same for every retriever. Also implemented: word and character TF-IDF (the character n-grams tolerate Kinyarwanda prefix changes), weighted hybrid fusion (default BM25 0.6 / TF-IDF 0.4, reciprocal-rank fusion optional) and the fine-tuned e5 embedding retriever.
- **Referral policy.** Rules run in order: low score, ambiguous top two (margin), classifier–retriever disagreement, passage not reviewed. The first to fire gives the reason. Threshold and margin come from a grid search on validation data that maximises the share answered while keeping wrong answers served at or below 5% (referral/tuning.py). For the fine-tuned retriever the notebook chose threshold 0.612, margin 0 and agreement on.
- **Fallbacks.** If text-to-speech fails, the written answer is still returned. Without a trained classifier file the factory trains a small SVM on the knowledge base. Speech is switched off by null objects that return HTTP 503 for voice questions.

3. Testing approaches and results

Approach	Why selected	How conducted	Results
Automated unit and integration tests	The pipeline swaps retrievers, classifiers and speech components, so each contract and every error path must keep working	pytest: 35 unit, 26 integration, 1 performance test. Heavy models replaced by fakes (tests/conftest.py). Cases: empty and over-long questions, script text in a question, invalid audio, over-size audio, speech disabled, missing TTS, unreviewed passage, off-topic question, API key checks	62 tests collected. The committed .coverage file records 88% of 1,043 statements; lowest are retrieval/embedding.py (0%), speech/stt.py (41%) and speech/tts.py (45%), which need model weights. ruff reports no issues
Held-out model evaluation	Retrieval and referral decide whether a farmer gets a wrong answer, so they are measured on unseen data	Notebook with "seen" splits (new questions on training passages) and "unseen" splits (passages held out); R@k, MRR, WER, macro-F1, bootstrap 95% intervals; 40 hand-written out-of-scope questions	Sections 3.2 and 3.3 (Tables 4 and 5)
Performance benchmark	Proposal sets a 5 s response target and a low-resource setting	scripts/benchmark.py: per-stage latency percentiles and throughput with 8 client threads	Section 3.4 (Table 6)

Table 2. Testing strategies used.

3.1 Testing with different inputs

Input	Expected behaviour (asserted in the tests)	Test
Valid Kinyarwanda question	HTTP 200, status answered from a reviewed passage; with audio on, a WAV (RIFF) is returned; feedback accepted with 201	test_api: text round trip
Empty, missing or 501-character question	HTTP 422; nothing is searched	test_api: empty and oversized
Off-topic question; text containing a script tag	Status referred with the officer-referral message; no crash, nothing executed	test_api, test_pipeline
Valid voice recording	Transcription returned in heard and the question answered	test_api: voice question

Input	Expected behaviour (asserted in the tests)	Test
Non-audio bytes; file over the size limit; clip over 30 s	HTTP 422 (invalid_audio); HTTP 413; clip rejected	test_api, test_audio
Recording transcribed as silence	Referred with reason speech_not_understood	test_pipeline
Speech or TTS not available	Voice question returns HTTP 503 (service_unavailable); a missing TTS still returns the written answer	test_api, test_pipeline
Passage not signed off by an officer	Never served; the question is referred	test_pipeline
Unknown answer id; rating of 9	HTTP 404; HTTP 422	test_api: feedback validation
Officer or admin call with no key, a wrong key, or no key configured	HTTP 401; HTTP 401; HTTP 503. With the right key: review 201 and stats returned	test_api: API key tests

Table 3. Input cases, expected behaviour and the tests that check them.

3.2 Retrieval and referral

Values are shown as seen / unseen test splits.

System (test split, seen / unseen)	R@1	R@5	MRR@10
BM25	0.487 / 0.489	0.740 / 0.726	0.595 / 0.599
TF-IDF (character 3–5)	0.513 / 0.568	0.806 / 0.851	0.637 / 0.687
multilingual-e5, off the shelf	0.400 / 0.316	0.619 / 0.502	0.491 / 0.396
BM25 + fine-tuned e5 (weight 2.0)	0.745 / 0.790	0.932 / 0.973	0.825 / 0.870
Fine-tuned e5 (adopted)	**0.893 / 0.866**	**0.990 / 0.988**	**0.936 / 0.923**

Table 4. Retrieval accuracy (results/retrieval_final_test_all_systems.csv).

Referral rule (test)	Answered	Right among answered	Wrong served
Seen: threshold only	0.936	0.910	0.084
Seen: + margin + agreement	0.819	0.933	0.055
Unseen: threshold only	0.903	0.882	0.106
Unseen: + margin + agreement	0.690	0.938	0.043

Table 5. Referral rules (notebook cell 15).

Fine-tuned e5 against BM25 on the test set: R@1 0.888 vs 0.488, a gain of +0.400 (95% interval 0.361 to 0.441).

Two cautions are reported in the notebook:

- 27% of seen-test questions are near-duplicates (cosine ≥ 0.85) of training questions; without them R@1 is 0.874.
- The unseen test covers only 20 passages (329 questions, R@1 interval 0.809 to 0.922).

The stricter referral rule cut wrong answers served from 8.4% to 5.5% (seen) and 10.6% to 4.3% (unseen), at the price of answering fewer questions. All 40 out-of-scope questions (not agriculture, medical, financial, legal, and farming topics outside the knowledge base) were referred; the highest out-of-scope score was 0.557, below the 0.612 threshold.

3.3 Classification and speech

- **Classification** (TF-IDF + logistic regression / SVM, test macro-F1): 0.894 / 0.933 on seen passages but 0.607 / 0.586 on unseen ones; accuracy 0.908 / 0.942 (seen) and 0.851 / 0.854 (unseen); about 0.2 ms per question. The labels are keyword drafts, not officer-corrected, which limits what these numbers show.
- **Speech-to-text:** fine-tuning lowered WER from 0.301 to 0.277 (gain +0.024, 95% interval +0.010 to +0.038 over 200 clips) and CER from 0.063 to 0.058. On synthetically degraded audio WER rose to 0.311 (8 kHz, 20 dB noise) and 0.404 (8 kHz, 10 dB); these are not real farmer recordings.
- **Text-to-speech:** each voice was spoken, transcribed by the ASR and compared with the text. The stock voice scored WER 0.454 and the fine-tuned voice 0.761 on 100 held-out sentences (real speech through the same ASR: 0.263), so the stock voice was kept.

3.4 Performance and compatibility

Benchmark file	Passages	Retriever	p50	p95	Throughput	Start-up
benchmark_hybrid_941.json	941	hybrid	46.0 ms	54.9 ms	43.1 req/s	18.8 s
benchmark_embedding_941.json	941	embedding	716 ms	1,050 ms	1.7 req/s	17.8 s
benchmark_laptop.json	12	hybrid	8.8 ms	60.1 ms	137.6 req/s	0.23 s
benchmark_light_60runs.json	12	hybrid	12.3 ms	52.0 ms	68.0 req/s	0.27 s

Table 6. Text-pipeline latency (Windows 10, Python 3.13.3, 4-core Intel CPU, 8 client threads, speech disabled).

Every run is far inside the 5 s target, apart from one 121 s outlier in the embedding run (cause not investigated). Model training and evaluation ran on a Tesla T4 GPU.

Not yet measured, and the main testing gaps: other browsers and phones, a low-end versus high-end comparison, memory use (peak_memory_mb is empty in the benchmark files) and STT/TTS latency inside the web app.

4. Code quality practices

The back end is about 1,650 lines in 13 packages under src/mfasha, with about 500 lines of tests. Practices in use:

- **Abstraction and polymorphism.** Four small interfaces (Retriever, Classifier, SpeechToText, TextToSpeech in interfaces.py). BM25, TF-IDF, hybrid and embedding retrievers are interchangeable, and a config value selects one.
- **Dependency inversion and modularity.** factory.py is the only place that names concrete classes; the pipeline, routes and tests receive collaborators through constructors, so tests build the whole app from fakes.
- **Single responsibility and open/closed.** Routes only validate, call the pipeline and log. The pipeline owns the order of stages and none of their logic. A new safety rule is one small class added to the referral chain (chain of responsibility) with no other change.
- **Encapsulation.** Frozen dataclasses for Passage, RetrievalHit and Settings; constructors validate their arguments (BM25 parameters, hybrid weights); a StageTimer context manager records timings without touching stage code; a repository class hides all database access.
- **Algorithms and problem solving.** Inverted-index BM25 with score normalisation, weighted and reciprocal-rank fusion, quantile grid search for the referral threshold, WAV decoding with mixing down and resampling, and character n-gram matching for Kinyarwanda word variants.
- **Safety and robustness.** Constant-time API key comparison, request size and duration limits, typed error classes mapped to HTTP codes 413, 422 and 503, and graceful degradation when speech components are missing.
- **Tooling and delivery.** pyproject.toml with ruff rules E, F, I, B, UP; a Makefile; environment-based settings; a Dockerfile with a non-root user and health check; docker-compose with PostgreSQL; a README with install, API and structure; results and benchmark files committed so figures can be traced.

5. Known limitations and next steps

Known limitations

- The pilot with farmers and officers, and the validation interviews, are not documented in the repository; they are the main outstanding item against the proposal.
- The default retriever is hybrid (BM25 + TF-IDF), whose accuracy is not in the results files, and its default threshold is 0.20. The reported referral figures (threshold 0.612, 40 of 40 out-of-scope referred) apply to the fine-tuned e5 retriever, which needs the model folder.
- All 941 passages were marked reviewed by one reviewer over four days; the notebook itself warns to keep that only if every passage was truly checked. Classifier labels still need officer correction.

- The read-aloud button asks the server again with audio enabled, so each use logs an extra interaction; the app also speaks every answer, whereas the notebook leaves answers containing digits or brackets as text (74% of passages) because the voice was trained on plain sentences.

Next steps

- Record native-speaker ratings.
- Benchmark STT/TTS latency and memory.
- Test more browsers and devices.